

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems
COURSE CODE: CSA0508

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity.*
(BL2, BL3)

Activity Tool : Experiments (High)
Assessment Tool Weightage: 40%

Dr.G.Ayyappan

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	For the relation STUDENT_COURSE(SID, SName, CID, CName, Instructor, Grade), identify all functional dependencies and determine the candidate keys.	BL2 – Understand	5
2	Demonstrate the process of converting an unnormalized relation into 1NF with step-by-step justification and example.	BL3 – Apply	5
3	Given relation R(A,B,C,D,E) with FDs: $A \rightarrow BC$, $BC \rightarrow D$, $D \rightarrow E$. Analyze and decompose it to 2NF eliminating partial dependencies.	BL3 – Apply	5
4	Apply 3NF decomposition on the relation EMPLOYEE(EmpID, EmpName, DeptID, DeptName, ManagerID) with transitive dependencies.	BL3 – Apply	5
5	Verify whether R(A,B,C,D) with FDs: $AB \rightarrow C$, $C \rightarrow D$, $D \rightarrow A$ is in BCNF. If not, decompose it into BCNF.	BL3 – Apply	5
6	Experimentally verify lossless-join decomposition for a given relation using the tabular (Chase) algorithm.	BL3 – Apply	5

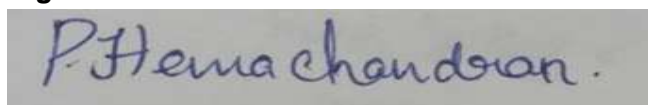
7	Identify the multi-valued dependencies in TEACHER(TID, Subject, Hobby) and decompose it into 4NF.	BL3 – Apply	5
8	Demonstrate how join dependencies lead to redundancy. Decompose a given relation to 5NF (Project-Join Normal Form).	BL3 – Apply	5
9	Given a poorly designed database schema for a college, identify all anomalies (insertion, deletion, update) and normalize it to 3NF.	BL2 – Understand	5
10	Compute the closure of attribute sets for the given FDs and determine all candidate keys for the relation.	BL3 – Apply	5

Code of Conduct:

I P. HEMACHANDRAN (192525440) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results

Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation
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1. STUDENT_COURSE – Functional Dependencies and Candidate Keys

Given:

STUDENT_COURSE(SID, SName, CID, CName, Instructor, Grade)

Assume:

- One student can take many courses.
- One course can have many students.
- Each course has one course name and instructor.
- A student's grade is specific to the student-course combination.

Step 1: Identify Functional Dependencies

The functional dependencies are:

1. **SID → SName**
2. **CID → CName**
3. **CID → Instructor**
4. **(SID, CID) → Grade**

Therefore:

CID → CName, Instructor

and

(SID, CID) → SName, CName, Instructor, Grade

Step 2: Find Candidate Key

Calculate closure of {SID, CID}:

(SID,CID)+

Initially:

{SID,CID}

Using SID → SName:

{SID,CID,SName}

Using CID → CName, Instructor:

{SID,CID,SName,CName,Instructor}

Using (SID,CID) $\rightarrow$ Grade:

{SID,CID,SName,CName,Instructor,Grade}

It contains all attributes.

Therefore:

Candidate Key=(SID,CID)

Answer

Functional Dependencies:

SID $\rightarrow$ SName CID $\rightarrow$ CName,Instructor (SID,CID) $\rightarrow$ Grade

Candidate Key:

(SID,CID)

2. Conversion of an Unnormalized Relation into 1NF

Consider:

Unnormalized relation

STUDENT(SID, SName, Phone, Course)

SID	SName	Phone	Course
101	Ram	9876543210, 9123456780	DBMS, OS
102	Shyam	9988776655	DBMS
103	Mohan	9123456789, 9988001122	OS, AI

The problem is that **Phone** and **Course** contain multiple values.

Step 1: Identify repeating groups

For student 101:

- Phone = 9876543210, 9123456780
- Course = DBMS, OS

These are not atomic values.

Step 2: Make values atomic

Each cell should contain only **one value**.

1NF Relation

STUDENT_1NF(SID, SName, Phone, Course)

SID	SName	Phone	Course
101	Ram	9876543210	DBMS
101	Ram	9123456780	DBMS
101	Ram	9876543210	OS
101	Ram	9123456780	OS
102	Shyam	9988776655	DBMS
103	Mohan	9123456789	OS
103	Mohan	9123456789	AI
103	Mohan	9988001122	OS
103	Mohan	9988001122	AI

Justification

A relation is in **1NF** when:

- Each attribute contains atomic values.
- There are no repeating groups.
- Each row represents a single record.

Therefore, the above relation satisfies **1NF**.

3. 2NF Decomposition

Given:

R(A,B,C,D,E)

FDs:

$A \rightarrow BC$ $BC \rightarrow D$ $D \rightarrow E$

Step 1: Find Candidate Key

Calculate:

A^+

Initially:

$A^+ = \{A\}$

Using $A \rightarrow BC$:

$A^+ = \{A, B, C\}$

Using $BC \rightarrow D$:

$A^+ = \{A, B, C, D\}$

Using $D \rightarrow E$:

$A^+ = \{A, B, C, D, E\}$

Therefore:

A is a candidate key

Step 2: Check partial dependency

A partial dependency occurs when a **non-prime attribute depends on part of a composite key**.

Here the candidate key is a **single attribute A**.

Therefore, there cannot be a partial dependency on a part of A.

Important observation

This relation is already in **2NF** provided it is in 1NF.

However, it has a **transitive dependency**:

$A \rightarrow D \rightarrow E$

This is a 3NF issue, not a 2NF issue.

2NF Result

$R(A, B, C, D, E)$ is already in 2NF

No decomposition is required **for 2NF**.

For 3NF, it can later be decomposed into:

$R1(A,B,C)$ $R2(B,C,D)$ $R3(D,E)$

4. 3NF Decomposition of EMPLOYEE

Given:

EMPLOYEE(EmpID, EmpName, DeptID, DeptName, ManagerID)

Assume the FDs:

$EmpID \rightarrow EmpName, DeptID, ManagerID$ $DeptID \rightarrow DeptName$ $ManagerID \rightarrow EmpName$

The important transitive dependency is:

$EmpID \rightarrow DeptID \rightarrow DeptName$

Step 1: Identify transitive dependency

We have:

$EmpID \rightarrow DeptID$

and:

$DeptID \rightarrow DeptName$

Therefore:

$EmpID \rightarrow DeptName$

through DeptID.

Step 2: Create Employee relation

EMPLOYEE(EmpID, EmpName, DeptID, ManagerID)

Primary Key:

EmpID

Step 3: Create Department relation

DEPARTMENT(DeptID, DeptName)

Primary Key:

DeptID

Final 3NF decomposition

Relation	Attributes	Primary Key
EMPLOYEE	EmpID, EmpName, DeptID, ManagerID	EmpID
DEPARTMENT	DeptID, DeptName	DeptID

Thus:

EMPLOYEE(EmpID,EmpName,DeptID,ManagerID) DEPARTMENT(DeptID,DeptName)

The transitive dependency is eliminated.

5. BCNF Verification and Decomposition

Given:

$R(A,B,C,D)$

FDs:

$AB \rightarrow C$ $C \rightarrow D$ $D \rightarrow A$

Step 1: Find candidate keys

AB^+

$AB^+ = \{A,B\}$

Using $AB \rightarrow C$:

$AB^+ = \{A,B,C\}$

Using $C \rightarrow D$:

$AB^+ = \{A,B,C,D\}$

So:

AB is a candidate key

BC^+

Start:

BC

$C \rightarrow D$:

B,C,D

$D \rightarrow A$:

A,B,C,D

Therefore:

BC is a candidate key

BD⁺

$D \rightarrow A$:

A,B,D

$AB \rightarrow C$:

A,B,C,D

Therefore:

BD is a candidate key

Step 2: Check BCNF

BCNF requires:

For every non-trivial FD $X \rightarrow Y$, X must be a superkey.

Check:

$AB \rightarrow C$

AB is a candidate key.

✓ Satisfies BCNF.

$C \rightarrow D$

C is not a superkey.

X Violates BCNF.

$D \rightarrow A$

D is not a superkey.

X Violates BCNF.

Therefore:

R is not in BCNF

Step 3: Decompose using $C \rightarrow D$

Create:

$R_1(C,D)$

and:

$R_2(A,B,C)$

Step 4: Check BCNF

For $R_1(C,D)$:

$C \rightarrow D$

C is the key.

Therefore R_1 is in BCNF.

For $R_2(A,B,C)$:

$AB \rightarrow C$

AB is the key.

Therefore R_2 is in BCNF.

Final answer

$R_1(C,D)$ $R_2(A,B,C)$

6. Lossless-Join Decomposition Using Chase Algorithm

Consider:

$R(A,B,C)$

FDs:

$A \rightarrow B$ $B \rightarrow C$

Decompose into:

$R_1(A,B)$ $R_2(B,C)$

Step 1: Create Chase Table

Row A B C

R1 a b c1

R2 a2 b c

Here:

- a, b, c represent distinguished symbols.
- Other symbols represent different values.

Step 2: Apply $A \rightarrow B$

The rows having the same A value must have the same B value.

After applying:

Row A B C

R1 a b c1

R2 a2 b c

Step 3: Apply $B \rightarrow C$

Both rows have the same B value b.

Therefore their C values must become the same.

Row A B C

R1 a b c

R2 a2 b c

A row now contains all distinguished symbols:

(a,b,c)

Conclusion

Therefore, the decomposition is **lossless**.

$\{R1(A,B), R2(B,C)\}$ is lossless

7. 4NF Decomposition of TEACHER

Given:

TEACHER(TID, Subject, Hobby)

Suppose a teacher can have:

- Multiple subjects
- Multiple hobbies

And subjects and hobbies are independent.

Therefore, the multivalued dependencies are:

$TID \twoheadrightarrow \text{Subject}$

and:

$TID \twoheadrightarrow \text{Hobby}$

Example

TID Subject Hobby

T01 DBMS Cricket

T01 DBMS Music

T01 OS Cricket

T01 OS Music

This creates unnecessary combinations.

Step 1: Decompose

Create:

TEACHER_SUBJECT(TID, Subject)

TID Subject

T01 DBMS

T01 OS

Create:

TEACHER_HOBBY(TID, Hobby)

TID Hobby

T01 Cricket

T01 Music

Step 2: Check 4NF

Both relations contain only independent multivalued information.

Therefore:

TEACHER_SUBJECT(TID,Subject) TEACHER_HOBBY(TID,Hobby)

are in **4NF**.

8. 5NF / Project-Join Normal Form

Consider:

R(Supplier,Part,Project)

Suppose supplier-part-project relationships can be reconstructed from three independent pairwise relationships.

There is a join dependency:

$\ast(\text{Supplier,Part}),(\text{Part,Project}),(\text{Supplier,Project})$

Example

Original relation:

Supplier Part Project

S1 P1 J1

S1 P2 J1

S2 P1 J2

S2 P2 J2

If the information can be represented through independent pairwise relationships, storing all combinations creates redundancy.

Step 1: Decompose

Create:

R1(Supplier,Part) R2(Part,Project) R3(Supplier,Project)

Step 2: Reconstruct

The original relation can be obtained through:

$R = R1 \bowtie R2 \bowtie R3$

Conclusion

If this decomposition is lossless and the join dependency is represented by the decomposed relations, the schema satisfies **5NF/PJNF**.

R1(Supplier,Part) R2(Part,Project) R3(Supplier,Project)

9. College Database – Anomalies and 3NF

Consider a poorly designed relation:

COLLEGE(StudentID, StudentName, CourseName, InstructorID, InstructorName, InstructorPhone)

Example:

StudentID	StudentName	Course	InstructorID	InstructorName	Phone
101	Arun	DBMS	I01	Kumar	98765
102	Ravi	DBMS	I01	Kumar	98765
103	Priya	OS	I02	Anitha	91234

A. Insertion Anomaly

Suppose we want to add a new course:

AI → Instructor I03

But there is no student enrolled yet.

We cannot insert the course information without creating a student record.

Therefore:

Insertion anomaly

B. Deletion Anomaly

Suppose student 103 is the only student taking OS.

If we delete student 103, we may also lose:

- OS course information
- Instructor I02 information

Therefore:

Deletion anomaly

C. Update Anomaly

Instructor I01's phone number occurs in multiple rows.

If the phone number changes, every row must be updated.

If one row is missed, inconsistent information occurs.

Therefore:

Update anomaly

Step 1: Identify dependencies

$\text{StudentID} \rightarrow \text{StudentName}$ $\text{CourseName} \rightarrow \text{InstructorID}$

$\text{InstructorID} \rightarrow \text{InstructorName, InstructorPhone}$

Step 2: Remove transitive dependency

Because:

$\text{CourseName} \rightarrow \text{InstructorID}$

and:

$\text{InstructorID} \rightarrow \text{InstructorName, InstructorPhone}$

we have:

$\text{CourseName} \rightarrow \text{InstructorName, InstructorPhone}$

Step 3: Decompose into 3NF

STUDENT

STUDENT(StudentID,StudentName)

COURSE

COURSE(CourseName,InstructorID)

INSTRUCTOR

INSTRUCTOR(InstructorID,InstructorName,InstructorPhone)

This removes the major insertion, deletion, and update anomalies.

10. Attribute Closure and Candidate Keys

Consider:

$R(A,B,C,D)$

with FDs:

$A \rightarrow B \quad B \rightarrow C \quad C \rightarrow D \quad D \rightarrow A$

We calculate the closure of each attribute.

Step 1: A^+

Start:

$A^+ = \{A\}$

Using $A \rightarrow B$:

$\{A,B\}$

Using $B \rightarrow C$:

$\{A,B,C\}$

Using $C \rightarrow D$:

$A^+ = \{A,B,C,D\}$

Therefore A is a candidate key.

Step 2: B^+

Start:

$B^+ = \{B\}$

$B \rightarrow C$:

$\{B, C\}$

$C \rightarrow D$:

$\{B, C, D\}$

$D \rightarrow A$:

$B^+ = \{A, B, C, D\}$

Therefore B is a candidate key.

Step 3: C^+

Start:

$C^+ = \{C\}$

$C \rightarrow D$:

$\{C, D\}$

$D \rightarrow A$:

$\{A, C, D\}$

$A \rightarrow B$:

$C^+ = \{A, B, C, D\}$

Therefore C is a candidate key.

Step 4: D^+

Start:

$D^+ = \{D\}$

$D \rightarrow A$:

$\{A, D\}$

$A \rightarrow B$:

$\{A, B, D\}$

$B \rightarrow C$:

$D^+ = \{A, B, C, D\}$

Therefore D is a candidate key.

Final Answer

Attribute Set Closure Candidate Key?

A+	ABCD	Yes
B+	ABCD	Yes
C+	ABCD	Yes
D+	ABCD	Yes

Therefore:

Candidate Keys = $\{A\}, \{B\}, \{C\}, \{D\}$

Quick Normalization Summary

Normal Form	Main Requirement	Problem Removed
1NF	Atomic values	Repeating groups
2NF	No partial dependency	Partial dependency
3NF	No transitive dependency	Transitive dependency
BCNF	Every determinant is a superkey	Stronger FD anomalies
4NF	No non-trivial MVD unless determinant is superkey	Multivalued dependency
5NF	No problematic join dependency	Join dependency redundancy